



Quiz 09.08 (Note: Paper Quiz. Nothing to upload. Click for Review Guide [Updated Sunday 2:15 PM]))

Completion requirements

Due: Monday, September 8, 2025, 10:20 AM

DSC 360 – Quiz 1 Review Guide

Updated 2025-09-07 at 2:15 PM (removed F1 definition, added a few additional things you should know)

Note: This review guide is intended to help you focus your study, but it does not include every possible question. Review your notes carefully.

Logistics

- Date: Monday, start of class
- Time Limit: 15 minutes
- Format: Short answer and definitions (no multiple choice)
- Closed book: No notes or “cheat sheets” may be used during the quiz
- Scope: First week (Labs 00.1–00.3, lectures, discussions)

Topics You Need to Know

Syllabus

- Make sure you have read the syllabus carefully. Be prepared for one or two questions related to the Syllabus.

Linux / Shell Basics

- What a terminal is (plain-language definition)
- Navigation: `cd`, `ls`, `mkdir`, `rmdir`
- File operations: `mv`, `cp`, `rm`
- Special paths: `~` (home directory), `..` (parent directory)
- Run a Python file from the command line:
`python3 hello.py`

- Measure runtime with the time command:
time python3 mini_sentiment.py
- Be prepared for a “move/copy” task (example paths: from ~/Downloads to ~/Desktop/dsc360/labs/lab01/). Example commands:
mv ~/Downloads/file.txt ~/Desktop/dsc360/labs/lab01/ cp
~/Downloads/file.txt ~/Desktop/dsc360/labs/lab01/ rm
~/Desktop/dsc360/labs/lab01/file.txt
- On the quiz or the test, be prepared to answer: **Why is it important to use directories to organize your work?** (In other words, why not just putting all of your files on the Desktop or in the Downloads directory?)

LLM Basics

- Know what **LLM** stands for (**Large Language Model**)
- High-level idea: an LLM predicts the next **token** (word or part of a word) in a sequence
- Model size labels (e.g., 270m, 1b, 4b, 12b) and the rough trade-off: smaller → faster, larger → often better quality but slower
- Note: Be sure to watch the film or read the transcript from the first day about the rise of **agents**. There may be a question or two from that.

Ollama and API Use

- Roles of components:
 - **daemon** = **server** process (runs in background)
 - **CLI** = **client** you type into in the terminal
 - Python **API** = programmatic interface from your code
- Know what CLI (Command Line Interface) and API (Application Programming Interface) stand for
- Chat request message shape (**JSON** list of {role, content} objects):
[{"role": "user", "content": "Write Hello World in Python"}]
- Response includes **metadata** and a **message**; the text you want is in:
response.message.content
- Importance of function outputs: helper functions (e.g., query_ollama) should return a string, not print; printing belongs in main()

Prompting & Normalization

- Why vague prompts fail; write strict prompts for labels (e.g., “Answer with ONE WORD: positive or negative.”)
- Output cleanup in Python:
 - `.strip()` to remove extra whitespace at the beginning and end

- `.lower()` to "fold" case—i.e., make all lower case
- Substring checks using the operators `in` and `not in`

Classification Metrics

- Make sure you understand what is meant by true positives, false positives, etc.:
 - TP (true positives): actual positive, predicted positive
 - TN (true negatives): actual negative, predicted negative
 - FP (false positives): actual negative, predicted positive
 - FN (false negatives): actual positive, predicted negative
- You should know what we mean by an **imbalanced dataset**
 - A dataset is **balanced** if the **support** for the positive and negative classes is the same (e.g., in a set with $N=100$ examples, 50 are positive and 50 are negative)
 - Conversely, a dataset with $N=100$ examples, with 90 negative and 10 positive is **imbalanced**.
 - Imbalanced datasets often create difficulties for us in machine learning.
- You should understand when the *F1 score* would be a better metric than *accuracy*
 - Example question: **Suppose you had a dataset with 90 positive examples and 10 negative examples. Explain why the accuracy score is not a useful metric when evaluating a model on this dataset.** (A model that always guessed positive would get 90% accuracy even though it always misclassifies negative examples.)
 - **What metric would be more useful in this case?** (F1 score)
 - Note: You will not be asked to *compute* the F1 score or other metrics on the quiz